

Retrieval Augmented Gen



AI



API



LLM



Hallucinations



Building your Retrieval-Augmented Generation (RAG) for Custom LLMs



Prashanth Basappa

Follow

2 min read · May 22, 2024



61



2





While recently playing with RAGs and understanding why they matter to provide custom data for your LLM, there seems to be plenty of options to enable that. Ragie, the thing I stumbled upon on a podcast seems to be doing just that.

Get Prashanth Basappa's stories in your inbox

Join Medium for free to get updates from this writer.

[Subscribe](#)☐ Remember me for faster sign in

LLMs on their own have limitations, when it comes to providing contextually accurate and up-to-date information. This is where Retrieval-Augmented Generation (RAG) comes into play. RAG combines the strengths of retrieval-based and generation-based approaches to enhance the capabilities of LLMs. RAG addresses the limitations by allowing LLMs to fetch relevant information from external databases, ensuring that the responses are accurate, current, and contextually relevant. This approach also reduces the chances of the model “hallucinating” or making up information. Let’s see how we can build this up quickly with a terminal (virtual python env) and a OpenAI subscription:

Building a Custom LLM with Ragie

Ragie is a powerful tool that simplifies the process of building a custom LLM using RAG. Here’s a step-by-step guide to help you get started:

Step 1: Set Up Your Environment

```
python
# Example setup
pip install openai
pip install llama-index
```

Step 2: Prepare Your Data

The first step in building a RAG system is to prepare your data. This involves collecting documents or data that you want your LLM to use as a reference.

```
python
# Load your documents
documents = ["user_manual_1.txt", "user_manual_2.txt"] # This is your custom data
```

Step 3: Create Embeddings

Next, convert your documents into embeddings. Embeddings are numerical representations of your text data that capture the context and meaning of the content.

```
python
from llama_index import SimpleDirectoryReader, GPTVectorStoreIndex

# Create embeddings
reader = SimpleDirectoryReader("path_to_your_documents")
index = GPTVectorStoreIndex.from_documents(reader.load_data())
```

Step 4: Build the RAG Pipeline

Now, set up the RAG pipeline. This involves creating a query engine that can retrieve relevant information from your embeddings and augment the LLM's responses

Open in app ↗

Sign up

Sign in

Medium

🔍 Search

✍ Write



```
python
# Initialize the query engine
query_engine = index.as_query_engine(similarity_top_k=3)
# Example query
query = "How do I reset my device?"
response = query_engine.query(query)
print(response)
```

Step 5: Integrate with Your LLM

Finally, integrate the retrieval component with your LLM to generate responses that are both contextually relevant and accurate.

```
python
import openai
# Function to generate response using RAG
def generate_response(query):
    retrieved_info = query_engine.query(query)
    prompt = f"Using the following information: {retrieved_info}, please answer
    response = openai.Completion.create(
        engine="text-davinci-003",
        prompt=prompt,
        max_tokens=150
    )
    return response.choices[0].text.strip()

# Example usage
query = "How do I reset my device?"
print(generate_response(query))
```

Conclusion

By following these steps, you can build a custom LLM that leverages the power of RAG to provide accurate, contextually relevant responses. This approach will help your LLM stay up to date with the latest information.

Think of it as a bridge between static knowledge and real-time information lookup.

Retrieval Augmented Gen

+

AI

+

API

+

LLM

+

Hallucinations

+

**Written by Prashanth Basappa**

277 followers · 112 following

Aspiring Storyteller and Critical Thinker 📍 SEATTLE

Follow

Responses (2)



Write a response

What are your thoughts?

**Tushar Wagh**

Oct 6, 2025



Very precise and simple example.

Thank you Prashanth.



1

[Reply](#)



Rahulkumarparida

Jan 23



Nice needed exactly this



[Reply](#)

More from Prashanth Basappa



Prashanth Basappa · Jul 16, 2021

The technology behind Apple's AirTag

My first time exploring the world of IoT and the technology the powers it. Playing around...



27



In CodeX by Prashanth Basappa · Apr 25, 2022

Software migrations and Best Practices

Best practices for executing migrations well, at both small and large scales.



7





Prashanth Basappa · Aug 30, 2022

Hiking the Enchantments trail

The hardest day hike I have done



155



Prashanth Basappa · May 13, 2024

Why is Spades card game so addictive?

Since March 30th, 2024, my weekends have been consumed by this card game that has...



See all from Prashanth Basappa

